

## Response to Referee #1

We thank the referee for the time devoted to reviewing our manuscript and for providing feedback that helps us clarify the scope, motivation, and interpretation of our results. While some concerns appear to stem from misunderstandings of the model structure and empirical exercises, we are grateful for the opportunity to strengthen the exposition. Below, we address each point in turn. For ease of reference, referee comments appear in italics.

### 1. Motivation for choosing Ecuador and comparison with other IMF programs

#### Referee comment:

*Please explain and justify why you choose Ecuador. In addition, please explain how the IMF program in Ecuador is different from IMF programs in other commodity-exporting countries, and how such differences affect your outcomes.*

#### Response:

Ecuador is selected for two key reasons. First, its 2020–25 IMF-supported program provides unusually detailed projections for *individual* fiscal instruments (labor-income taxes, capital-income taxes, transfers, public investment, and public consumption) as well as commodity revenue. This level of granularity is essential for calibrating a realistic consolidation path and is rarely available for other commodity exporters.

Second, Ecuador is a fully *dollarized* and mostly externally indebted commodity exporter with historically volatile sovereign spreads. This makes it an ideal environment for studying how commodity-revenue fluctuations interact with fiscal consolidation and external borrowing costs. In contrast, many other commodity exporters operate with structural-balance rules or sovereign wealth funds, implying very different fiscal dynamics. We now highlight these differences explicitly in the revised text.

We have added the following paragraph in the introduction:

Ecuador provides a particularly informative environment because the 2020–2025 program specifies detailed paths for multiple fiscal instruments together with oil-revenue projections, allowing the program to be mapped into explicit policy paths in the model. More broadly, the paper’s qualitative mechanism applies to other commodity exporters that consolidate under external borrowing constraints and volatile sovereign spreads: consolidation performance is highly contingent on commodity-revenue realizations and the financing conditions they induce.

### 2. Foreign versus domestic debt

#### Referee comment:

*Only foreign debt is included in the fiscal policy block. Why not include domestic debt? Ecuador issues domestic debt, and the substitution margin is important.*

**Response:**

We thank the referee for raising this point. Ecuador indeed issues both external and domestic public debt. In the model we focus on external debt because the sovereign spread — which is a key transmission channel in our framework — is determined in international capital markets and reflects the country’s external borrowing conditions. Domestic public debt in Ecuador is largely priced off the sovereign spread observed in international markets and therefore inherits similar financing conditions.

In standard small open economy DSGE models, domestic and external public liabilities are often aggregated into a single debt instrument when capital markets are integrated. An alternative formulation would be to assume that a fixed fraction,  $\alpha \in [0, 1]$ , of government liabilities is held abroad and the remaining  $1 - \alpha$  domestically. Under perfect capital mobility and identical pricing conditions, such a decomposition is largely notational and does not alter the equilibrium dynamics of fiscal consolidation or the interaction between debt accumulation and the sovereign risk premium.

To clarify this interpretation, we have added the following footnote 9 in the model section explaining that the debt variable can be interpreted as the marginal borrowing instrument relevant for sovereign spreads and that the results would be unchanged under a standard  $\alpha$ -share decomposition of domestic and external debt.

The model represents government liabilities as a single debt instrument priced at the external interest rate plus a sovereign risk premium. In practice, a country can issue both domestic and external public debt. A standard alternative formulation in small open economy DSGE models assumes that a fixed fraction  $\alpha \in (0, 1)$  of public liabilities is held abroad while the remaining share is held domestically. Under integrated capital markets and common pricing conditions, this decomposition is largely notational and does not affect the equilibrium dynamics of fiscal consolidation or the link between public debt and sovereign spreads. We therefore interpret the government debt variable in the model,  $B_t^*$ , as one of the marginal borrowing instruments relevant for determining the country’s risk premium. As a reference, external debt represented around 70 % of Ecuador’s public debt during the sample period.

### 3. Interpretation of technology shocks in the historical decomposition

**Referee comment:**

*Figure 1 shows most of GDP is driven by technology shocks. What exactly are these shocks? Please elaborate on events such as the 2009 crisis, 2012, and after 2017.*

**Response:**

We appreciate the referee’s request for clarification. In the model, the “technology shock” corresponds to a neutral productivity disturbance affecting non-oil output. Importantly, this process includes a permanent component that governs the long-run growth rate of output.

The prominence of this shock in the historical decomposition reflects the model’s need to account for persistent changes in output growth during the sample period. In particular, the decline in non-oil output growth following the collapse in oil prices after 2014 represents a structural change

that stationary shocks in the model (such as fiscal, demand, or risk-premium shocks) cannot generate. The estimation therefore attributes this shift in trend growth primarily to the permanent productivity process.

We emphasize that these shocks should not be interpreted as literal technological innovations. Rather, they capture broader movements in underlying productive capacity that are not explicitly modeled, such as changes in the scale of economic activity associated with the commodity cycle or persistent adjustments in productivity following large demand shocks. The revised manuscript clarifies this interpretation in the discussion of the historical decomposition, as follows on page 16:

Looking at the non-oil GDP growth rate through the lens of the model points to technology shocks (dark blue bars) as the most important drivers. This is particularly relevant in the last part of the sample, in which permanent productivity shocks ( $a_t$ ) are the only mechanism through which the model can rationalize the persistent decline in output growth. Because the productivity process contains a permanent component that determines the long-run growth rate of non-oil output, the estimation attributes structural changes in trend growth to this disturbance. Other shocks in the model — including fiscal, risk-premium, and demand shocks — are stationary and therefore cannot generate permanent shifts in the growth rate of output.

A deeper structural explanation of this slowdown is beyond the scope of this paper. One possible interpretation is that growth patterns featuring weak (strong) demand induce a slowdown (rise) in productivity, as argued by Stadler (1990), Anzoategui et al. (2019), Jordà, Singh and Taylor (2020), and Furlanetto et al. (2025), among others.

We also added the following footnote 22 in the historical decomposition section:

Historical decompositions in DSGE models should therefore be interpreted as statistical allocations of fluctuations across structural disturbances rather than literal identifications of specific historical events.

## 4. Contribution of foreign variables

### Referee comment:

*Foreign shocks have very small impacts in Figure 1. This is counter-intuitive for a small open economy.*

### Response:

We thank the referee for raising this point. The historical decomposition reflects the relative importance of shocks given their estimated variances rather than their structural role in the model. In our estimation, fluctuations in Ecuador’s macroeconomic variables during the sample period are primarily associated with commodity-price movements, fiscal shocks, and sovereign risk-premium disturbances, while innovations to foreign output are estimated to have comparatively small variances.

This result is consistent with the macroeconomic experience of Ecuador and other commodity-exporting economies, where the terms of trade and domestic fiscal conditions often dominate the business cycle. In particular, the large oil-price fluctuations experienced during the sample period and the associated fiscal adjustments play a central role in driving domestic activity.

More broadly, the DSGE literature frequently finds that domestic shocks account for the majority of cyclical fluctuations even in small open economies once models are estimated on the data. For example, Medina and Soto (2016) and Melina, Yang and Zanna (2016) find that commodity-price and fiscal shocks dominate the historical decomposition of output in resource-rich economies. We have added the following footnote 24 in the discussion of the historical decomposition clarifying this interpretation.

It is worth noting that the relatively small contribution of foreign shocks reflects their estimated historical variance rather than their theoretical importance in the model. During the sample period, Ecuador’s business cycle was dominated by large oil-price fluctuations, fiscal adjustments, and movements in the sovereign risk premium. As a result, the estimation attributes most of the variation in domestic output to these shocks rather than to innovations in foreign output. Similar findings are common in estimated DSGE models of commodity-exporting economies, where terms-of-trade and domestic fiscal shocks typically account for a large share of macroeconomic fluctuations (see Medina and Soto, 2016; Melina, Yang and Zanna, 2016, for instance).

## 5. Monetary policy treatment

### Referee comment:

*How is monetary policy treated? Which panel of Figure 3 corresponds to monetary policy?*

### Response:

We thank the referee for this question. The model abstracts from an explicit monetary-policy block and is formulated in real terms, focusing on fiscal policy, external borrowing, and the sovereign risk premium as the key transmission mechanisms. This choice reflects the paper’s objective of analyzing fiscal consolidation in an externally indebted small open economy, where financial conditions and debt dynamics play a central role.

Ecuador’s full dollarization since 2000 provides a particularly transparent environment for this abstraction, as there is no independent domestic policy rate. In the model, the relevant interest rate is given by the exogenous world interest rate plus a sovereign risk premium that depends on external indebtedness.

Because the framework does not include a monetary-policy rule, there are no monetary-policy shocks or impulse responses, and therefore no corresponding panel in Figure 3. We note, however, that the mechanisms we emphasize—namely the interaction between fiscal policy, external debt, and borrowing costs—are not specific to dollarized economies and would also operate in environments with an active monetary authority.

## 6. Persistence of tax-rate shocks

### Referee comment:

*It is odd to postulate tax rates following an  $AR(1)$  coefficient of 0.999. Tax rates should be exogenous policy variables.*

**Response:**

We thank the referee for pointing this out. In the previous version of the paper, the highly persistent tax rate processes were used as a technical device to implement the fiscal paths of the IMF program within a stochastic simulation framework, not in the estimation of the model or its solution where they were constant. Specifically, the near-unit-root processes allowed us to use sequences of shocks to keep the tax instruments close to the program targets over the consolidation horizon.

In the revised version of the paper, we have changed the implementation of the fiscal consolidation experiment and now solve the model under perfect foresight. Fiscal instruments, including tax rates, are imposed directly as deterministic paths corresponding to the IMF program. As a result, the highly persistent tax processes (with persistence coefficient 0.99) are only required after the consolidation window so that the simulation can be characterized by a quasi permanent increase in tax rates without changing the model's steady state. This revision addresses the referee's concern and simplifies the interpretation of the fiscal consolidation simulations.

## 7. Hump-shaped government investment response

**Referee comment:**

*Why is government investment hump-shaped in Figure 4?*

**Response:**

We thank the referee for this question. In the revised version of the paper, government investment does not display a hump-shaped response. Instead, it follows the path specified in the IMF-supported consolidation program, which features a front-loaded reduction in public investment followed by stabilization at a lower level. Because the model is now solved under perfect foresight, fiscal instruments are imposed directly as deterministic policy paths during the consolidation horizon. The dynamics shown in Figure 4 therefore reflect the program's policy trajectory. We have clarified this point in the discussion of the figure on page 24.

## 8. Exploding confidence bands for optimizer consumption

**Referee comment:**

*Why does the confidence interval of optimizer consumption explode over time?*

**Response:**

We thank the referee for this question. The trajectories shown in Figure 4 are not confidence intervals. Instead, the three lines correspond to deterministic simulations of the fiscal consolidation program under alternative oil-revenue paths (baseline, higher-than-expected revenues, and lower-than-expected revenues). The divergence across trajectories therefore reflects differences in the assumed commodity-revenue scenarios rather than statistical uncertainty around a single forecast.

In the revised manuscript we clarify this point in the description of Figure 4. We also report credible intervals for the baseline transition paths in Appendix J, which explicitly quantify the uncertainty arising from posterior parameter draws.

## 9. Restricted hours

### Referee comment:

*Why are the three lines for restricted hours the same? What is restricted hours?*

### Response:

We thank the referee for raising this question. "Restricted hours" correspond to the labor supply of hand-to-mouth households in the model. In the fiscal consolidation experiment, the trajectories of the fiscal instruments (tax rates, government consumption, government investment, and transfers) are identical across the alternative oil-revenue scenarios. These policy paths are imposed in levels following the IMF-supported program.

As a result, the contemporaneous income of hand-to-mouth households changes very little across scenarios, since the variables that directly affect their labor-supply decision—wages, taxes, and transfers—remain nearly identical. The alternative oil-revenue paths mainly affect GDP through the external sector and the sovereign risk premium, which changes fiscal variables when expressed as ratios to GDP but not their levels.

We note that this point was already discussed in the current footnote 30 in the manuscript.

## 10. Contributions and novelty

### Referee comment:

*The conclusion does not show any new findings. Are there new contributions? If not, it will be difficult to publish this manuscript.*

### Response:

We thank the referee for raising this point. In the revised manuscript we have rewritten the introduction and conclusion to clarify the paper's contribution relative to the existing literature on fiscal policy in commodity-exporting small open economies.

The paper makes three main contributions.

First, we study a fiscal consolidation that uses multiple fiscal instruments (taxes, transfers, public investment, and public consumption) following the trajectories specified in an IMF-supported program. Embedding such a policy package in a structural macroeconomic model allows us to analyze its macroeconomic implications in a coherent general-equilibrium framework.

Second, we show how different realizations of commodity revenues can generate very different macroeconomic outcomes even when fiscal policy follows exactly the same path. In the model, alternative oil-revenue trajectories primarily affect debt dynamics and sovereign spreads, which in turn alter the evolution of output, investment, and the external balance during the consolidation.

Third, the model incorporates heterogeneous households, allowing us to trace how fiscal consolidation affects different groups through distinct channels. Hand-to-mouth households react mainly through changes in transfers and labor income, while optimizing households adjust consumption and investment intertemporally through the risk-premium channel.

These contributions are now stated more clearly in the introduction and conclusion of the revised manuscript.

## Closing remark

We thank the referee for the suggestions, which have helped us clarify and improve the presentation of the model, results, and contributions. In the revised manuscript we have clarified several aspects of the empirical and institutional context, strengthened the discussion of the paper's contributions, and improved the exposition of the simulation results. We have also updated the fiscal consolidation experiment by implementing the program paths under perfect foresight, which allows agents to anticipate the policy measures and preserves the nonlinear dynamics of the model. Overall, these revisions enhance both the transparency of the analysis and the interpretation of the results.

Sincerely,  
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## References

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